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APPLIED RESEARCH

A Framework for Enablement of Augmented Reality in Railway Maintenance

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ABSTRACT Railway maintenance relies on effective assessment of asset health to ensure safety and reliability. It requires appropriate technologies, methodologies, and tools for monitoring the condition of railway assets. These technologies and tools often utilize advanced sensor technology for data acquisition, distributed computing for data processing, enhanced analytics for knowledge extraction, and visualization technologies for data representation. While significant progress has been made in data acquisition, processing, and analytics, data visualization has not received sufficient attention. In parallel, the emerging Industry 5.0 concept emphasizes human-centricity, resilience, and sustainability in industrial systems, highlighting the need for enhanced human-system interaction (HSI) in maintenance processes. HSI can be enhanced by using perception-enhancing technology such as Augmented Reality (AR). However, the effective integration of AR into railway maintenance requires structured frameworks that address the challenges of task workflow, interaction, and information presentation. Hence, the purpose of this paper is to propose a framework for enabling AR-based assistance in railway maintenance. Following a Design Science Research (DSR) methodology, the framework defines functional elements and information flows to support the execution of inspection tasks and is realized through an AR-based system that integrates procedural guidance, flexible workflow navigation, multimodal interaction, contextual information access, and automated inspection log generation. A controlled user study compares the AR-assisted inspection approach with a traditional inspection method using standardized usability measures. The results indicate that both approaches scored within the marginal usability range. These findings indicate the feasibility of AR-based assistance as a human-centric complement to existing railway maintenance practices, while motivating further refinement and evaluation in operational settings.

INDEX TERMS Augmented reality, decision support, human-centered maintenance, industry 5.0, railway maintenance, usability evaluation.

I. INTRODUCTION

Railway maintenance comprises technical, administrative, and managerial actions performed across the asset life cycle to retain or restore required functionality [1]. Railway assets, including railway tracks, sleepers, ballast, wheelsets, bogies, and axles, deteriorate due to factors such as mechanical load, exposure to the climate, and the environment [2], and material fatigue over time [3]. Effective railway maintenance aims to restore these assets as close as possible to an as-good-as-new (AGAN) condition, thereby preserving functional performance and supporting future asset life [4], [5]. From a maintenance engineering perspective, this relates to maintenance efficiency, which reflects how accurately inspection and maintenance actions are executed in accordance with prescribed procedures, acceptance limits, and documentation requirements [6]. In practice, achieving these objectives increasingly depends on how asset condition information is acquired, processed, visualized, and interpreted by human operators during maintenance task execution.

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Among the various maintenance activities, inspection tasks play a central role in achieving maintenance efficiency, as they directly influence how maintenance actions are initiated, prioritized, and verified across different railway assets [7]. In this study, railway wheelsets are used as a representative case for inspection, given their safety-critical role and well-defined inspection standards. Achieving maintenance efficiency, therefore, requires inspection tasks to be performed in a planned, timely, and procedurally consistent manner [8].

Despite these advances, the effectiveness of railway maintenance increasingly depends on how asset condition information is presented to and interpreted by human operators. Although significant digital advancements have been made in railway maintenance, including onboard diagnostics [9], Internet of Things (IoT) based sensing systems [10], digital twins (DT) [11], and predictive maintenance [12], maintenance and inspection tasks still rely heavily on human expertise and manual judgment. Maintenance technicians are often required to assess asset conditions and make safety-critical decisions manually [13]. They must consider data from various sources, such as visual inspection data, procedural documentation, and standards, to determine appropriate measures to be taken to proceed with the maintenance actions. Studies highlight that the effectiveness of maintenance actions strongly depends on how technicians understand, interpret, and execute tasks in real-world contexts [13], [14]. While these technologies have significantly improved data acquisition, processing, and predictive capabilities, less attention has been paid to how this information is visualized and integrated into technicians' real-time workflows. As a result, human-system interaction (HSI) during inspection and maintenance execution remains a critical challenge, particularly in safety-critical contexts.

However, performing these tasks is cognitively challenging for the technicians because the maintenance workflows are complex [15]. Technicians must follow established standards, complete multi-step procedures, consult relevant documentation, and handle physical tools while continually monitoring their surroundings to ensure safety [16]. This increases their cognitive load and affects the efficiency of the maintenance task. This leads to difficulties in decision-making and heightened risk of human error in critical maintenance operations [16].

At the same time, an emerging skills and experience gap across the railway industry makes maintenance execution even more difficult [17]. Senior and experienced technicians are retiring, while newer staff often lack experience with complex inspection and maintenance tasks. Recent research in human-centric maintenance highlights the need for digital tools that support decision-making, knowledge retention, and operator empowerment to mitigate this gap [18]. In this context, Industry 5.0 provides relevant paradigmatic framing. Unlike Industry 4.0, which prioritizes automation and data-driven efficiency, Industry 5.0 explicitly focuses on human-centricity, resilience, and sustainability as core design principles [19]. It highlights the need for technologies that complement and augment human skills rather than replace human workers, which has direct implications for how maintenance support tools should be designed. Consequently, there is a growing need for maintenance support solutions that improve human perception, understanding, and interaction with asset condition information during task execution.

Augmented Reality (AR), as a perception-enhancing technology, has gained attention for its ability to deliver hands-free, context-aware, real-time guidance directly augmented onto the physical environment [20], [21]. AR has the potential to support task execution, improve comprehension, and facilitate access to relevant information in industrial maintenance settings [22]. Within the railway sector, AR has been explored for inspection, remote collaboration, and visualization of maintenance procedures [23]. This work focuses on AR due to its suitability for supporting real-world inspection tasks. Unlike Virtual Reality (VR), which immerses users in a fully virtual environment and is primarily suited for training and simulation, AR enables contextual information to be overlaid directly onto physical assets while maintaining situational awareness, which is essential for safety-critical maintenance activities.

Despite the suitability of AR for such tasks, current AR-based solutions in railway maintenance focus primarily on instruction delivery rather than addressing the broader human-centered needs associated with complex maintenance tasks. They often focus on technology-driven instruction delivery and provide limited flexibility or contextual explanation tailored to technicians' experience or needs [24]. They usually lack the ability to pause, revisit, or navigate through the steps. Likewise, current AR tools rarely include explainable reasoning or decision-support capabilities that help technicians understand why an action is required or how it should be performed [22]. Such limitations highlight that AR maintenance systems should evolve into a more intelligent and transparent assistant rather than merely a digital overlay.

Addressing these challenges requires a structured approach that goes beyond individual system features and explicitly links inspection task demands to human-centric support mechanisms. To address these gaps, this study proposes a human-centric AR maintenance assistance framework, grounded in the context of Industry 5.0, and demonstrates its applicability through railway wheel inspection tasks. This is developed using a Design Science Research (DSR) approach. The framework formalizes how key inspection task demands, such as procedurally constrained workflows, high cognitive load, knowledge-intensive decision-making, and documentation and traceability requirements, can be systematically mapped to AR-based support mechanisms. Unlike most of the existing AR-based maintenance systems, which primarily address instruction delivery or isolated system features, the proposed framework operates at a higher level of functional abstraction. It defines reusable functional elements that are independent of specific AR hardware or implementation technologies, and models the information flows between these elements, capturing how

task state, user interactions, and decision-support outputs are exchanged during inspection execution. Furthermore, the framework introduces structured human-system collaboration mechanisms, including user-controlled workflow navigation, contextual explanations, and assistive decision support.

A controlled usability study involving 19 participants evaluates the framework through its instantiation by comparing the AR-assisted inspection approach with a traditional inspection method using the System Usability Scale (SUS). This evaluation provides empirical evidence of the framework's usability characteristics and its potential to support human-centric railway inspection tasks.

The main contributions of this paper are,

- A human-centric AR maintenance assistance framework grounded in Industry 5.0 principles. The framework defines technology-independent functional elements and explicit information flows and introduces structured human-system collaboration mechanisms that position AR as a collaborative assistant rather than a prescriptive tool.
- An AR-based instantiation of the proposed framework for railway wheel inspection. It demonstrates how inspection task demands can be translated into concrete AR-based mechanisms for interaction, contextual decision support, and documentation.
- An empirical usability evaluation of the framework instantiation through a controlled within-subjects study comparing the AR-assisted inspection approach with a traditional inspection method, providing a usability baseline and insights into user perceptions of AR-supported inspection workflows.

II. RELATED WORK

Research on digitalization in maintenance, AR, and human-centricity has expanded significantly, especially with the introduction of Industry 5.0. However, these research streams remain fragmented. With respect to railway maintenance, AR-enabled assistance and human-centric design are often studied in isolation, which motivates the gap of integrating these aspects for an Industry 5.0 appropriate human-centered AR-based maintenance assistant, proposed in this study.

A. DIGITALIZATION IN RAILWAY MAINTENANCE

Digitalization in Railway Maintenance has transformed how the infrastructure and rolling stock are being maintained. Traditional reactive and time-based strategies are being complemented by data-driven predictive and prescriptive approaches [25]. Condition monitoring systems that collect real-time data on asset behavior enhance situational awareness and support more informed decision-making. For example, research on performance indicators shows how degradation metrics and health indicators can be systematically used to prioritize maintenance actions and assess railway network performance over time [2]. Research demonstrates how onboard systems can detect early faults in freight vehicles by leveraging multi-sensor data streams, facilitating proactive maintenance scheduling [9]. IoT technologies have further driven digital adoption in the railway industry. They enable remote condition monitoring and cloud-based processing at scale and demonstrate the potential of how real-time asset data can trigger maintenance alerts before critical thresholds are exceeded [10]. DT technologies extend such monitoring capabilities by creating virtual replicas of physical assets that integrate historical and real-time information to support forecasting and decision support [11], [12]. Predictive analytics powered by machine learning (ML) models complements these digitalization efforts by extracting patterns from large datasets and estimating the remaining useful life (RUL) of components [26].

B. IMMERSIVE TECHNOLOGY APPLICATIONS IN THE RAILWAY INDUSTRY

Immersive technologies such as AR, Virtual Reality (VR), and Mixed Reality (MR) are increasingly being explored in the railway industry to make inspection and maintenance tasks more intuitive and safer for technicians. For example, AR/VR visualizations help inspectors interpret track environments more clearly and recognize safety risks that are difficult to perceive through conventional methods [23]. Research demonstrates how overlaying digital instructions onto real components can help technicians follow procedures more confidently and reduce ambiguity during tasks [24]. VR-based simulators are now being explored for train-driver training and safety-critical scenario simulations, with research highlighting that repeated simulator practice can significantly reduce driving errors [27]. Similarly, VR training environments have been developed for freight-wagon inspection, where learners can learn to practice inspection and maintenance procedures based on regulatory documentation [28]. Immersive technologies are also being explored for their support in railway maintenance and infrastructure monitoring, especially in providing immersive visualizations of infrastructure data, aiding defect detection, inspection procedures, and maintenance planning to enhance overall safety [23].

Table 1 summarizes representative studies identified within the scope of this review, which focused on AR, VR, and mixed reality applications in the railway maintenance and inspection domain. The comparison characterizes these studies across four dimensions: interaction modality, workflow flexibility, depth of decision support, and degree of human-factor consideration. These observations, drawn from studies within the defined scope of this review, motivate the need for a framework that explicitly addresses these dimensions in an integrated and human-centric manner.

C. HUMAN-CENTRIC AND DECISION SUPPORT SYSTEMS IN RAILWAY MAINTENANCE

With Industry 5.0, digital technologies are considered as tools that augment, rather than replace, human expertise. Human-centric and decision-support approaches in maintenance put worker well-being, resilience, and human-system

TABLE 1. Comparison of representative immersive technology studies in railway and industrial maintenance contexts within the scope of this review.

Study and Application Context	Domain	Technology and Interaction Modality	Workflow Flexibility	Decision Support	Human-Factor Consideration
AR assembly guidance for Industry 4.0 operators [20], [21]	Manufacturing	AR with Visual overlay	Rigid, sequential	Instruction delivery	Low - Moderate
AR knowledge management in industrial maintenance [22]	Industrial maintenance,	AR with Visual overlay	Structured, rigid	Knowledge retrieval	Moderate
AR for failure analysis in rolling stock [17]	Railway	AR with Visual overlay	Not defined	Failure analysis support	Low - Moderate
Human-centric maintenance with AR, AI and speech [18]	Railway	AR with Multimodality (speech and gesture)	Partially flexible	AI-assisted guidance	Moderate
AR/VR for railway infrastructure monitoring [23]	Railway	AR/VR with Visual overlays	Not defined	Visualization only	Low
AR for railway maintenance operations [24]	Railway	AR with Visual overlay	Rigid, sequential	Instruction delivery	Low, procedure-driven
Train-driver training and freight-wagon inspection [27]	Railway	VR with Immersive simulation	Scenario-based	None	Moderate, training focus

interaction at the center, arguing that predictive and diagnostic systems must enhance human judgement instead of operating as opaque closed boxes [16]. Early railway maintenance decision-support systems focused on using Reliability, Availability, Maintainability, Safety (RAMS) and life-cycle cost (LCC) analysis to help infrastructure managers choose cost-effective maintenance policies, but primarily at a planning level and with limited attention to technicians' cognitive demands [2]. More recent research emphasizes that human and organizational factors must be considered throughout the system design and development lifecycle, including maintenance task design, interfaces, and procedures [29], [30]. Research on decision-support models for rail maintenance control rooms shows how operators integrate condition data, work orders, and operational constraints, and demonstrates that poorly designed tools and procedures can lead to overload, loss of situation awareness, and suboptimal maintenance decisions [31], [32]. Such challenges are also present during inspection tasks like railway wheel inspection, where technicians must combine visual observations, procedural documentation, acceptance criteria, and operational constraints to make safety-critical decisions. These findings highlight a broader Human-System Interaction (HSI) perspective, which advocates designing support systems around actual cognitive tasks, information flows, and task conditions that might increase the likelihood of human error encountered by railway maintenance personnel.

D. RESEARCH GAP SYNTHESIS

The three research streams reviewed above highlight a structural gap at their intersection. Condition monitoring, IoT-based sensing, and predictive analytics have matured considerably and provide reliable data acquisition and decision support at the management and planning level. However, a gap remains in how this information reaches on-site technicians during task execution in a form that is actionable, cognitively comfortable, and integrated within procedural workflows. Existing AR-based systems in the railway domain partially address this gap through instruction delivery, but they operate primarily at a feature level, providing visual overlays or step-by-step guidance without flexible workflow control, contextual explanations tailored to technicians' decision-making needs, or structured human-system collaboration mechanisms. This study addresses the intersection of these gaps by proposing a framework that operationalizes human-centric AR assistance as a structured set of functional elements and information flows, grounded in Industry 5.0 principles and evaluated through a railway wheel inspection case.

III. MATERIAL AND METHODS

This study follows the DSR methodology [29]. The methodological framework follows the interaction between environment, design, and knowledge base, as illustrated in Figure 1, ensuring both practical relevance to railway maintenance and methodological rigor grounded in established research foundations.

In terms of the DSS framework, the environment layer captures the real-world context of the railway case study and maps the inspection task demands addressed in this work. These demands were identified through background study and literature review. They include procedurally constrained, multi-step inspection workflows and high cognitive load during task execution. In addition, inspection tasks rely heavily on domain knowledge and standards to support decision-making. Finally, extensive documentation requirements must be satisfied throughout the inspection process.

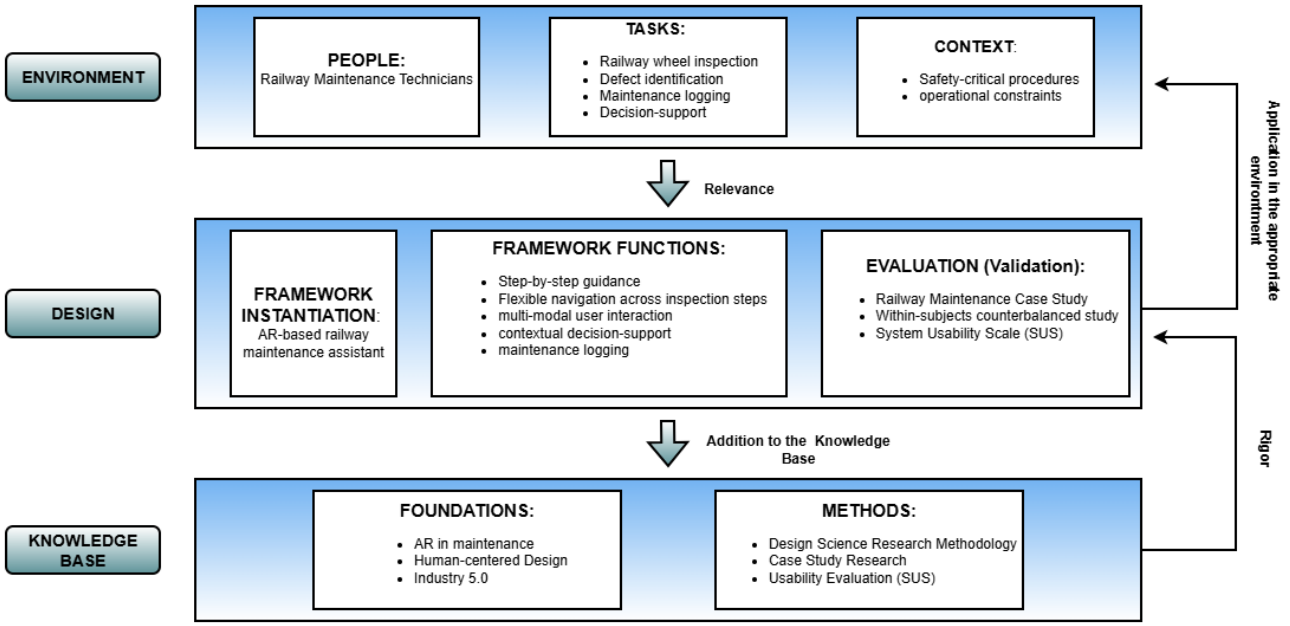


FIGURE 1. DSR framework illustrating the relationship between the railway maintenance environment, the development of a human-centric AR maintenance assistance framework, and its instantiation as an AR-based maintenance assistant, and the underlying knowledge base informing the study.

The design layer represents two of the core contributions of this work, in which a human-centric AR maintenance assistance framework for railway inspection tasks is developed. The framework is instantiated as an AR-based maintenance assistant for railway wheel inspection and evaluated through a controlled user study. A single-case study of railway maintenance, specifically focusing on wheel inspection tasks, is presented. The assistant provides step-by-step inspection guidance while allowing flexible navigation across inspection steps, supports multimodal interaction through speech and gestures, and enables contextual ''how'' and ''why'' explanations, defect logging, and automated documentation. The artifact is demonstrated and evaluated through a railway maintenance case study using a within-subjects, counterbalanced experimental design, in which usability is assessed using the SUS score. The evaluation outcomes inform iterative refinement of the system, ensuring alignment with user needs.

The knowledge base layer grounds the research in established foundations, including augmented reality in maintenance, human-centered design, and Industry 5.0 principles, as well as validated methodological approaches such as DSR, case study research, and usability evaluation. The directional links between the layers emphasize both relevance to practice and rigor in research.

Following the DSR process model [29], [30], the research activities in this study correspond to six phases (Figure 2). The Problem Identification and Motivation phase is addressed through a background study and literature review that identifies the inspection task demands, like procedurally constrained workflows, high cognitive load, knowledge-intensive decision-making, and documentation requirements, that the artifact is designed to address. The Define Objectives phase translates the identified task demands into framework requirements, establishing what the artifact must support. The Design and Development phase encompasses the proposed human-centric AR maintenance assistance framework, which constitutes the primary design artifact of this study. The Demonstration phase is realized through the instantiation of the framework as a functional AR-based maintenance assistant for railway wheel inspection, showing how the framework can be applied in a concrete maintenance context. The Evaluation phase is conducted through a controlled within-subjects usability study comparing the AR-based inspection approach with a traditional inspection method. The Communication phase is represented by the dissemination in a scientific publication. The primary contribution of this work is a design artifact, i.e., the proposed framework, complemented by evaluative knowledge derived from the usability study.

The subsections below describe the AR system architecture, the implemented use-case, participant characteristics, experimental procedure, data analysis, and ethical considerations.

A. FRAMEWORK INSTANTIATION AND SYSTEM ARCHITECTURE

The proposed human-centric AR maintenance assistance framework was instantiated as a functional AR-based maintenance assistant through substantial technical design and development effort. The instantiation required the integration of multimodal interaction, workflow control, contextual

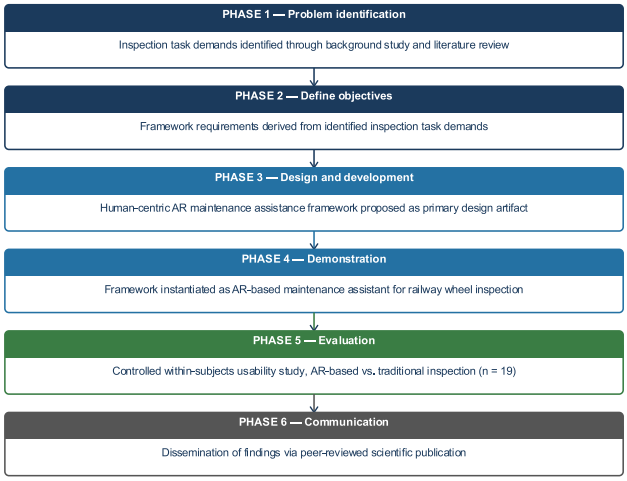


FIGURE 2. Mapping of research activities to the DSR process model phases [30]; (n = number of participants in usability study).

decision support, and inspection data management within a real-time AR environment. The system was developed for the Microsoft HoloLens 2 using Unity 2022.3.62f3 LTS and the Microsoft Mixed Reality Toolkit (MRTK), which provides native support for spatial interaction, hand tracking, and speech input. HoloLens 2 is a lightweight device and leaves both hands-free during task execution, which is an important consideration for physical inspection tasks. Participants wore the device for approximately 20-25 minutes per session, with most reporting no discomfort. However, some participants noted mild discomfort towards the end of the session, suggesting that prolonged wear across full maintenance shifts may present ergonomic challenges. The device provides two to three hours of active use on a single charge. However, extended use across full maintenance shifts, the impact of variable lighting conditions typical of outdoor or depot-based railway sites, and speech recognition performance under high ambient noise levels were not evaluated and constitute practical deployment considerations for future field evaluations.

1) DESIGN RATIONALE

The design rationale reflects how the identified inspection task demands are translated into framework requirements and subsequently realized in the system instantiation. Railway inspection tasks are typically performed in environments where technicians must handle physical tools, consider the surrounding environment, and adhere to strict safety procedures. Traditional inspection methods often rely on paper-based checklists or handheld devices, which require frequent shifts of attention and manual documentation. The proposed system was therefore designed to deliver guidance directly within the inspector's field of view while preserving flexibility and user control.

Key design requirements included:

(i) Support for step-by-step procedural guidance without enforcing a rigid sequence.

(ii) Multimodal interaction through speech and gestures.

(iii) Contextual explanations to support understanding and decision-making; and

(iv) Real-time defect logging with automatic documentation.

(v) User-controlled management of AR interface elements, enabling panels to be repositioned or temporarily minimized.

2) TECHNICAL ARCHITECTURE

Technical architecture represents the technical realization of the proposed framework instantiation and is organized into three main layers: an AR interaction layer, an application logic layer, and a data and logging layer. From a technical perspective, the primary challenge was to realize the framework's functional elements within a responsive, real-time AR system while maintaining flexibility, user control, and robustness during inspection execution. This required explicit separation of interaction handling, workflow logic, and data management, as well as careful coordination between these components to support event-driven user interaction. Figure 3 shows the layered architecture showing multimodal user interaction through HoloLens 2, workflow management via the application logic layer, and structured inspection data handling and summary generation in the data and logging layer.

The AR interaction layer takes care of rendering holographic content and processing user input. The system is deployed on the Microsoft HoloLens 2, with application development done using Unity and MRTK. MRTK provides native support for spatial interaction, hand tracking, and speech recognition, which provides multi-modal interaction in maintenance environments. Instruction panels, buttons, and informational overlays are spatially anchored to the users' head movement to maintain stability during movement. Visual feedback, such as highlights, confirmations, and overlays, is provided for all user actions to confirm selections and state changes.

The application logic layer implements the core technical mechanisms required to support flexible inspection execution. At its core, the Workflow Controller is implemented as a state machine that maintains task state and governs transitions between inspection steps in response to user actions. The state machine operates over a defined state set S, as given in eq (1):

$$S = \left\{ \begin{array}{c} \mathit{Initializing}, \mathit{StepActive}\,(i)\,, \mathit{HowDisplayed}\,(i)\,, \\ \mathit{WhyDisplayed}\,(i)\,, \mathit{DefectLogging}\,(i)\,, \\ \mathit{Paused}, \mathit{Completed} \end{array} \right\} \tag{1}$$

where $i \in \{1, \ldots, N\}$ denotes the index of the current inspection step, and N is the total number of steps in the workflow. The system begins in the Initializing state and transitions to StepActive(1) upon loading the inspection procedure. From StepActive(i), all subsequent transitions are governed by the user-triggered transition function δ(StepActive(i), e), where e denotes the triggering event,

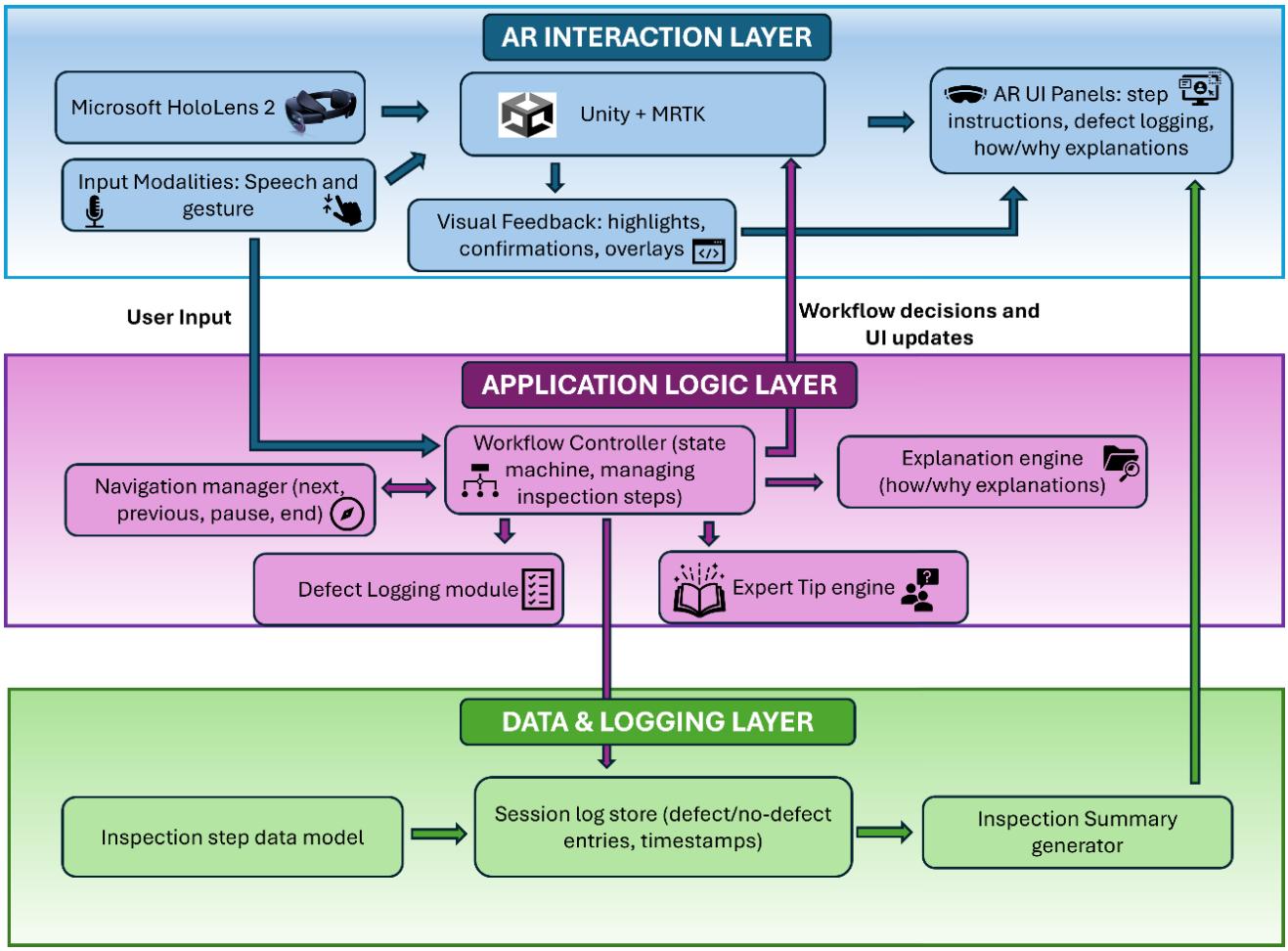


FIGURE 3. Layered system architecture of the AR-based railway maintenance assistant.

as defined in (2):

$$\delta\left(StepActive\left(i\right), e\right) = \left\{\begin{array}{l} HowDisplayed\left(i\right), if\ e = How \\ WhyDisplayed\left(i\right), if\ e = Why \\ DefectLogging\left(i\right), if\ e = Defect \\ StepActive\left(i+1\right), if\ e = Next, i < N \\ Completed, if\ e = Next, i = N\ or\ e = End \\ StepActive\left(i-1\right), if\ e = Back, i > 1 \\ Paused, if\ e = Pause \end{array}\right\} \quad (2)$$

When a defect is logged, the transition to DefectLogging(i) additionally triggers the Expert Tip Engine to present context-based recommendations before returning to the active step. All transient display and logging states return to StepActive(i) upon displaying the contextual information.

This design enables sequential progression while allowing user-controlled navigation, such as enabling users to move forward or backward through the inspection steps and to pause or resume the workflow as needed. The Navigation Manager interfaces with the workflow controller to handle navigation-related actions, ensuring that user commands are mapped to valid step transitions and bounded within the defined workflow. The Explanation Engine retrieves and presents contextual 'how' and 'why' information associated with each inspection step, supporting user understanding and decision-making. This content is pre-authored and stored within the Inspection Step Data Model, keyed by step identifier. 'How' content provides procedural guidance describing how the inspection action for that step should be executed. 'Why' content provides the underlying rationale and acceptance criteria grounded in the relevant inspection standards. This pre-authored approach was deliberately chosen over dynamically generated content to ensure the standard, reliability, and consistency of explanations in a safety-critical inspection context.

Defect-related actions are handled by the Defect Logging Module, which records defect and no-defect entries during inspection. When a defect is logged, the Expert Tip Engine is triggered to provide context-based recommendations, highlighting related components that may require further inspection. These recommendations are generated based on predefined mappings between inspection steps and expert knowledge.

The Data and Logging Layer takes care of the structured representation and storage of inspection information. The Inspection Step Data Model defines the properties of each inspection step, including instructional content, explanatory information, and defect-related options. During runtime, user actions and inspection outcomes are recorded in the Session Log Store, which captures defect and no-defect entries together with system-generated, session-based timestamps for each inspection step. After an inspection session, the Inspection Summary Generator compiles the recorded data into a structured summary. This summary is presented to the user within the AR interface and serves as a digital inspection record that can be reviewed or exported for further processing.

3) INSPECTION AND INTERACTION WORKFLOW

Algorithm 1 gives a summary of the event-driven control logic that governs the run-time behavior of the AR assistant application. The workflow starts with the inspection procedure, presents step-specific instructions, processes user interactions, records inspection outcomes, and generates an automatic maintenance log summary upon completion.

The workflow supports step-by-step progression with user-controlled navigation, allowing inspectors to move forward or backward through the procedure and to pause or resume the task as needed, without enforcing a rigid execution sequence.

Algorithm 1 Event-Driven Workflow of the AR-Based Railway Inspection System

```
Start application
Load inspection workflow (steps, how/why info, expert tips)
Initialise AR UI (instruction panel + controls)
Enable multimodal input (speech + gestures)
Set currentStep = 1
Set sessionActive = true

while sessionActive do
    Display(currentStep)
    Wait for user action

    if user requests "How" then
        ShowHow(currentStep)
    else if user requests "Why" then
        ShowWhy(currentStep)
    else if user logs defect then
        Log(currentStep, "defect")
        ShowExpertTip(currentStep)
    else if user logs no-defect then
        Log(currentStep, "no defect")
    else if user selects Next then
        currentStep = min(currentStep + 1, totalSteps)
    else if user selects Back then
        currentStep = max(currentStep − 1, 1)
    else if user selects Pause then
        Pause until Resume
    else if user selects End then
        sessionActive = false
    end if
end while

GenerateSummaryLog()
DisplaySummary()
End session
```

B. USER STUDY DESIGN

To evaluate the usability and practical applicability of the proposed human-centric AR maintenance assistance framework through its instantiation, a structured user study was conducted. The study assessed how the AR-based system supports railway inspection tasks in comparison to a traditional inspection approach. The evaluation focused on perceived usability, interaction flow, and user experience under controlled conditions, while reflecting realistic maintenance workflows. A within-subjects counterbalanced experimental design was employed to enable direct comparison between the two inspection approaches and to minimize potential learning and order effects.

1) PARTICIPANTS

A total of nineteen participants took part in the user study. The participant group consisted primarily of early-career railway maintenance technicians, representing a realistic population of potential end users for AR-based maintenance systems. This participant profile was intentionally selected, as early-career personnel are likely to benefit most from digital assistance tools that support procedural execution, knowledge transfer, and decision-support during inspection tasks.

Participants had varying levels of experience with railway maintenance activities. Some participants reported prior exposure to inspection procedures through training or academic coursework, while others had limited hands-on experience. In contrast, most participants had little to no prior experience with head-mounted AR devices, making them suitable for evaluating the learnability and usability of the system for first-time or novice AR users. This diversity in experience allowed the study to capture a broad range of user perspectives and reduce bias toward AR familiarity.

Participation was voluntary, and no incentives were offered. All participants were able to complete the study within a single session. Participants ranged in age from 19 to 39 years ($M = 26.9$, $SD = 5.8$). The group comprised 14 men and 5 women. All participants reported normal or corrected-to-normal vision. Seven participants used vision correction during the study, such as contact lenses or vision spectacles, without reported interference.

2) EXPERIMENTAL DESIGN

The study employed a within-subjects counterbalanced experimental design, in which each participant completed the same railway wheel inspection task using two different inspection approaches:

(i) A traditional inspection method, and

(ii) The proposed AR-based inspection system.

Figure 4 shows the railway wheel used for rim measurement in the case study.

In the traditional inspection condition, participants were provided with a printed inspection checklist and procedure document corresponding to the railway wheel rim inspection



FIGURE 4. The railway wheel rim region is considered during the inspection task used in the case study.

task. The document presented the same inspection steps covered in the AR-based condition, formatted as a numbered list with brief textual descriptions. No contextual explanations, visual guidance, or decision-support assistance was provided. Participants performed the physical inspection on the same railway wheel used in the AR condition, consulting the printed document as needed, and recorded inspection outcomes, defect or no-defect, manually on a provided paper form. This condition is representative of common current practice in railway maintenance environments, where technicians typically rely on paper-based procedures and manual documentation during inspection tasks.

A within-subjects design was selected to allow direct comparison of usability perceptions between the two approaches while controlling individual differences such as experience level, inspection speed, and decision-making style. To mitigate learning, fatigue, and order effects, the sequence in which participants performed the two inspection conditions was counterbalanced. Participants were assigned to one of two groups based on a simple numbering scheme: one group performed the traditional inspection first, followed by the AR-based inspection, while the other group completed the AR-based inspection first and then the traditional inspection.

This counterbalancing strategy ensured that neither inspection approach systematically benefited from prior task familiarity. Each participant completed the study individually and in isolation, and participants were instructed not to discuss their experiences or opinions with others until the completion of all study sessions. This measure was taken to reduce the influence of peer feedback and to ensure independent usability assessments.

3) PROCEDURE

Each study session followed a structured, three-phase procedure consisting of an introductory briefing, task execution, and post-task evaluation.

a: INTRODUCTORY BRIEFING

At the beginning of the session, participants were welcomed and provided with a brief overview of the study objectives and the inspection task. They were informed that the study aimed to evaluate inspection approaches rather than individual performance. Participants were explicitly told that they could withdraw from the study at any time without providing a reason. Written informed consent was obtained before proceeding.

For the AR-based condition, participants received a short introduction to the head-mounted display, including how to wear and adjust the device, basic interaction techniques, and available navigation options. This introduction focused on enabling safe and comfortable interaction and did not include task-specific training beyond what was necessary to operate the system.

b: TASK EXECUTION

Participants completed the railway wheel inspection task using both inspection approaches. In the traditional condition, participants followed conventional inspection procedures without AR support, relying on standard guidance and personal judgment. In the AR-based condition, participants used the proposed system to perform the same inspection workflow, receiving step-by-step guidance, contextual explanations, and interaction support through speech and gesture input.

During both conditions, participants completed the task independently under supervision. The supervisor ensured that participants followed the assigned procedure and did not receive external assistance. The duration of the AR-based inspection phase was approximately 20-25 minutes per participant, while the traditional inspection phase varied slightly depending on individual pace.

c: POST-TASK EVALUATION

After completing each inspection condition, participants filled out a usability questionnaire referring specifically to the inspection approach they had just completed. This process was repeated after both conditions, resulting in two independent usability assessments per participant.

TABLE 2. SUS questionnaire used in the study.

ID	Statement
Q1	I think that I would like to use this system frequently.
Q2	I found the system unnecessarily complex.
Q3	I thought the system was easy to use.
Q4	I think that I would need the support of a technical person to be able to use this system.
Q5	I found the various functions in this system were well integrated.
Q6	I thought there was too much inconsistency in this system.
Q7	I would imagine that most people would learn to use this system very quickly.
Q8	I found the system very cumbersome to use.
Q9	I felt very confident using the system.
Q10	I needed to learn a lot of things before I could get going with this system.

4) USABILITY MEASURES AND DATA ANALYSIS

Usability was assessed using the SUS, a standardized and widely validated questionnaire for evaluating perceived usability of interactive systems [31]. SUS consists of ten

statements, alternating between positively and negatively worded items, each rated on a five-point Likert scale ranging from Strongly disagree (1) to Strongly agree (5). The standard SUS items used in this study are listed in Table 2. The instrument produces a single usability score, ranging from 0 to 100, that reflects overall system usability and allows comparison across different systems and interaction approaches. Prior research [32], [33] has shown that a SUS score of approximately 68 represents average or acceptable usability. Scores in the range of 50–68 are commonly classified as marginal usability, suggesting that a system is usable but may require improvement, whereas scores below 50 are generally considered as poor usability. In contrast, SUS scores above 80 mean excellent usability and high user acceptance. This method was selected due to its reliability, simplicity, and suitability for comparing different interaction approaches.

The SUS score for each participant was computed following the standard scoring procedure.

$$Score_i = \begin{cases} R_i - 1, & if\ i \in \{1, 3, 5, 7, 9\} \\ 5 - R_i, & if\ i \in \{2, 4, 6, 8, 10\} \end{cases} \quad (3)$$

Let R_i denote the raw response to an item on a five-point Likert scale. For positively worded items (items 1, 3, 5, 7, and 9), the adjusted score was calculated as $R_i - 1$, whereas for negatively worded items (items 2, 4, 6, 8, and 10), the adjusted score was calculated as $5 - R_i$. The adjusted item scores $Score_i$ were then summed and multiplied by 2.5 to obtain the final SUS score, as defined in eq (3) - (4). The resulting score ranges from 0 to 100, with higher values indicating better perceived usability.

Mathematically, the SUS score is computed as:

$$SUS = 2.5 \times \sum_{i=1}^{10} S_i \quad (4)$$

where $Score_i$ represents the adjusted score of item i after applying the positive or negative item correction.

Each participant completed the SUS questionnaire twice: once after completing the traditional inspection and once after completing the AR-based inspection. The same set of SUS questions was used for both conditions, with wording adapted only to refer to the respective inspection approach. This enabled direct within-subject comparison of perceived usability between the two methods.

SUS scores were calculated according to standard scoring procedures. Descriptive statistics, including mean scores and standard deviations, were computed separately for the traditional and AR-based inspection conditions. Given the within-subjects design, comparative analysis focused on identifying differences in perceived usability between the two approaches. In addition to quantitative scores, informal qualitative feedback and observations were recorded to capture user impressions related to interaction flow, clarity of guidance, and perceived cognitive effort.

5) ETHICAL CONSIDERATIONS

The user study involved human participants and was conducted in accordance with institutional guidelines for low-risk research. All participants voluntarily agreed to take part in the survey. Participants were informed about the study objectives, procedures, and their right to withdraw at any time if they wished to opt out.

No personally identifiable information was collected, and all responses were anonymized. The study posed minimal risk to participants and did not involve any form of deception or physical intervention. Special attention was given to participants' comfort and safety during the use of the head-mounted AR device.

IV. RESULTS

This section presents the results of the work from two complementary perspectives. First, the proposed human-centric AR maintenance assistance framework for railway inspection tasks is presented as the primary research outcome. Its realization through an AR-based system instantiation is also described. Second, the framework is validated through a usability evaluation. The AR-based inspection approach is compared with a traditional inspection method.

A. HUMAN-CENTRIC AR MAINTENANCE ASSISTANCE FRAMEWORK

The primary result of this research is the proposed human-centric AR maintenance assistance framework for railway inspection tasks. The framework defines a set of interconnected functional elements and explicit information flows that collectively support inspection workflow execution, human-system interaction, contextual decision support, and documentation and traceability during safety-critical railway inspection tasks. Figure 5 illustrates the proposed framework as a set of four interconnected functional elements, each characterized by defined inputs and outputs and linked through information flows during inspection execution. The framework is grounded in Industry 5.0 principles and emphasizes human-centric support, positioning AR as an assistive technology that augments, rather than replaces, human decision-making.

Each element in the proposed framework has clearly defined responsibilities and interacts with the others through explicit inputs and outputs, forming a closed-loop assistive system grounded in Industry 5.0 principles.

At the core of the framework is the Inspection Workflow Management element, which represents the procedural backbone of the inspection task. This element receives inspection procedures and task definitions derived from standards and inspection guidelines. It is responsible for maintaining the current inspection step and the overall task state. In this framework, task state refers to the contextual representation of the inspection progress, including the active inspection step, its completion status, and the inspectors' position within the overall procedure. The task state does not represent inspection decisions or outcomes; rather, it provides the procedural context required to synchronize interaction,

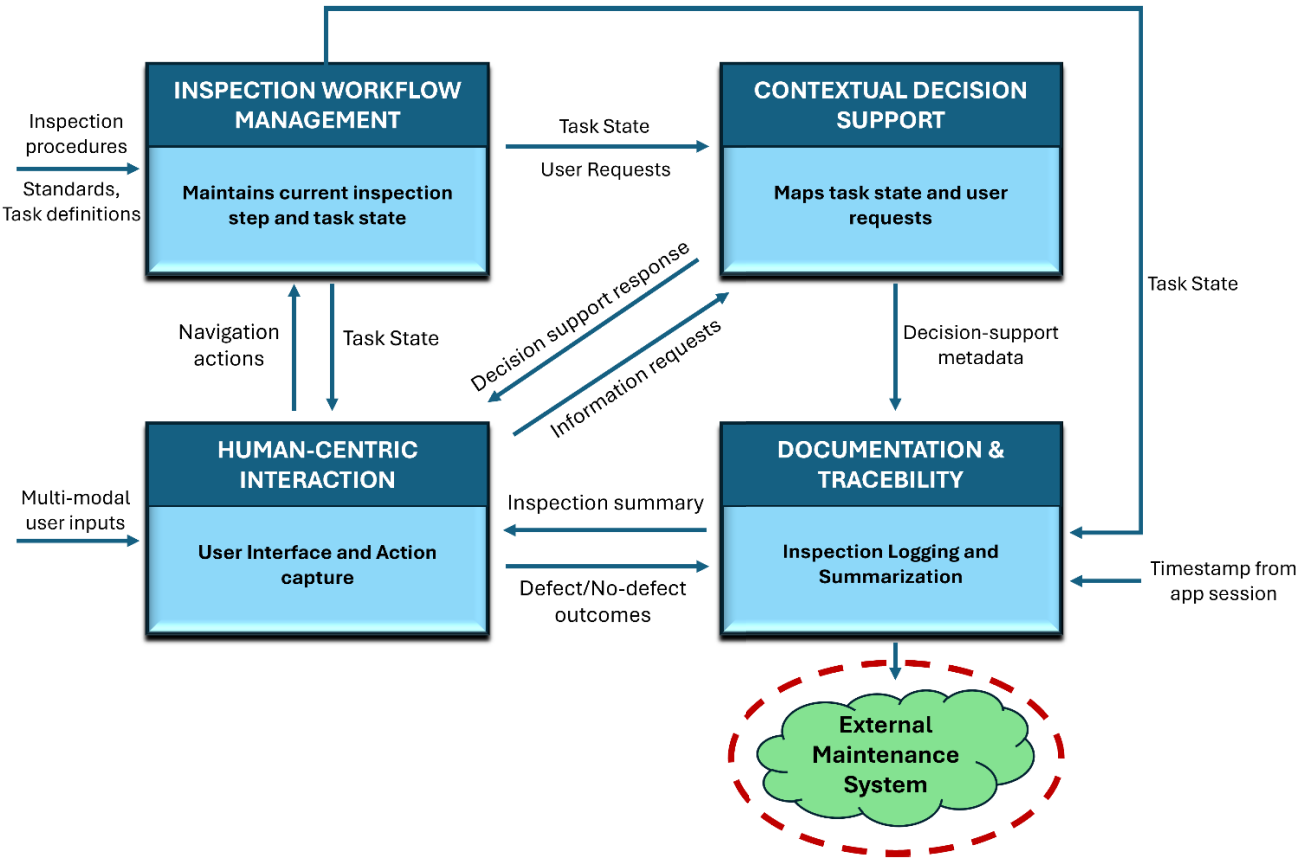


FIGURE 5. Human-centric AR maintenance assistance framework for railway inspection tasks, showing the functional elements and information flows supporting workflow management, interaction, decision support, and documentation.

decision support, and documentation. The workflow element outputs the current task state to other framework elements to ensure consistent and coordinated system behavior.

The Human-Centric Interaction element serves as the primary interface between the user and the framework. It captures user input through multi-modal interactions such as speech and gestures, enabling user actions such as workflow navigation, information requests, and defect or no-defect logging. Based on the task state received from the workflow element, this component determines which interaction options and interface elements are available at a given moment. The interaction element forwards navigation actions to the workflow management element, forwards information requests and defect-related actions to the decision support element, and forwards inspection outcomes to the documentation element. Importantly, this element does not contain procedural logic or decision-support intelligence; its role is to capture and communicate human actions rather than interpret them.

The Contextual Decision Support element provides assistive intelligence during the execution of inspections. It receives the current task state from the workflow management element, together with user requests and information requests originating from the interaction element. Using this information, it generates contextual explanatory content tailored to the active inspection step. This may include explanations that describe how an inspection action should be performed, or explanations that convey the underlying rationale and acceptance criteria. This may also include domain expertise and provide recommendations triggered when defects are logged, highlighting related components or follow-up inspection considerations. The decision support element outputs this explanatory information exclusively to the interaction element for presentation to the user.

The Documentation and Traceability element is responsible for the persistent recording and summarization of inspection activities. It receives inspection outcomes and defect/no-defect actions from the interaction element, together with task state from the workflow management element and system-generated, session-based timestamps. In addition, this element can capture inspection-related metadata, such as step identifiers, inspection context, and auxiliary information relevant for traceability. These inputs are used to create structured inspection records that associate each inspection step with its outcome, timing, and procedural context. Upon completion of the inspection session, this element generates an automated inspection summary. During inspection execution, the documentation element remains passive and does not influence workflow progression or decision-support. As illustrated in Figure 5 the framework might further allow

optional integration with external maintenance systems. This is considered a potential extension for future work.

Overall, the proposed framework explicitly separates procedural control, human interaction, decision support, and documentation into distinct functional elements. By defining clear responsibility boundaries and information flows, the framework provides a structured and reusable basis for designing human-centric AR maintenance assistance systems for railway inspection tasks. In this study, the framework is instantiated as an AR-based maintenance assistant for railway wheel inspection and subsequently validated through a controlled usability evaluation.

Table 3 illustrates how the proposed framework operationalizes human-centric support by mapping the identified inspection task demands into corresponding AR-based support mechanisms. By mapping this relationship, the framework provides a structured and transparent basis for supporting safety-critical railway inspection tasks while accounting for procedural, cognitive, and documentation-related challenges.

TABLE 3. Mapping of identified inspection task demands to framework support mechanisms.

Inspection Task Demand	Framework Support Mechanism
Multi-step and procedurally constrained inspection workflows	Structured inspection workflow with step-by-step guidance and flexible navigation
High cognitive load during task execution	Contextual step-by-step guidance presented directly in the field of view
Knowledge gaps related to inspection actions and acceptance criteria	On-demand explanations embedded within the inspection workflow
Decision uncertainty when identifying and interpreting defects	Assistive decision-support mechanisms triggered by inspection outcomes
Documentation burden and traceability requirements	Integrated defect logging and generation of structured inspection summaries

To demonstrate how the proposed framework can be realized in practice, it is instantiated as an AR-based maintenance assistant for railway wheel inspection. The following subsection describes the system architecture and interaction design that implement the framework's functional elements and information flows.

B. AR-BASED MAINTENANCE ASSISTANT FOR RAILWAY WHEEL INSPECTION

This subsection presents the AR-based maintenance assistant and its system architecture, which instantiates the proposed human-centric AR maintenance assistance framework. The architectural layers of the AR-based system collectively implement the functional elements defined by the proposed framework. Specifically, the application logic layer realizes the inspection workflow structure and task state management, the AR interaction layer supports human-centric interaction through multimodal input and visual feedback, the explanation and expert-tip components implement contextual decision support, and the data and logging layer realizes documentation and traceability through structured inspection records and session-based timestamps.

The application implements step-by-step inspection guidance directly within the user's field of view and integrates multimodal interaction through speech commands and gesture-based input. The developed system allows users to progress through inspection steps at their own pace, revisit previous steps, pause the workflow, and manage the visibility of interface elements to maintain situational awareness.

The application provides decision support through contextual explanations, allowing users to request ''how'' and ''why'' information for each inspection step, thereby enabling a deeper understanding of both task execution and the underlying rationale. During inspection, users can log defects or confirm the absence of defects directly within the AR interface. When a defect is reported, the system presents expert-tip recommendations, highlighting related components that may require additional inspection. Upon completing the workflow, the application automatically generates a structured inspection summary, thereby reducing manual documentation effort.

The implemented system demonstrates the technical feasibility of instantiating the proposed framework within a real-time AR environment, integrating workflow control, multimodal interaction, contextual decision support, and structured inspection logging within a single system.

The snapshots in Figure 6 illustrate the implemented AR-based railway maintenance assistant as evaluated in the user study. Unlike the traditional inspection method, which relies on a printed checklist, manual defect recording, and provides no contextual guidance or decision support, the AR system delivers each of these capabilities within a single integrated workflow.

Figure 6(a) shows step-by-step inspection guidance presented through spatially anchored AR instruction panels, enabling inspectors to follow the inspection workflow directly within their field of view. As shown in Figure 6(b), inspectors can log defects during task execution; when a defect is reported, the system provides expert-tip recommendations to support further inspection and decision-making. Upon user request, contextual 'how' or 'why' explanations are displayed, Figure 6(c) shows a 'how' explanation, offering additional guidance on inspection actions and supporting task understanding without interrupting the workflow. After the inspection process, the system automatically generates a structured maintenance log summary (Figure 6(d)), compiling recorded defect and no-defect entries and reducing the need for manual documentation.

Together, these capabilities illustrate how the proposed framework operationalizes human-centric AR assistance and highlight the gap between the AR-assisted and traditional inspection approaches.

C. USABILITY EVALUATION

The proposed framework is validated through a usability evaluation of its instantiation as an AR-based maintenance

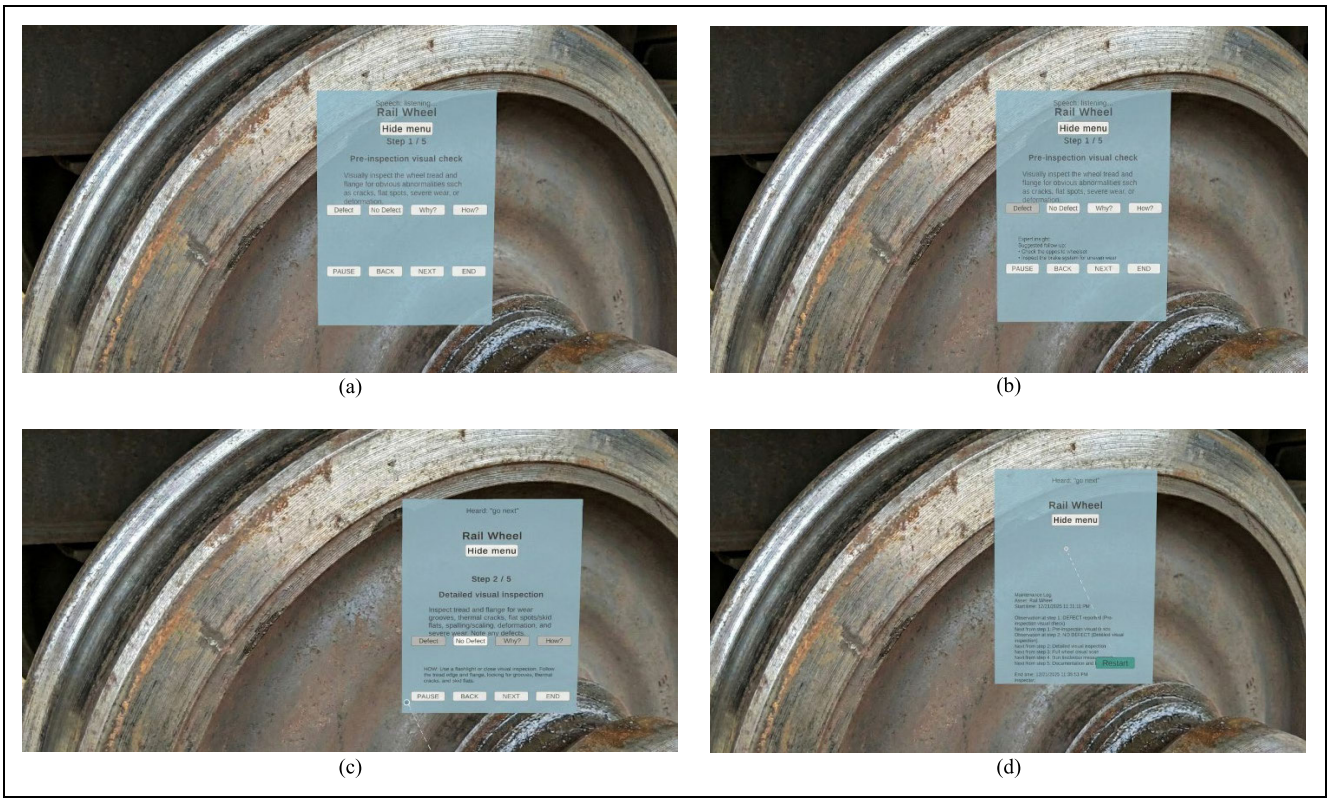


FIGURE 6. AR-based railway maintenance assistant: (a) step-by-step inspection guidance, (b) defect logging with expert tips, (c) contextual "how" explanations, and (d) automatically generated maintenance log summary.

assistant for railway wheel inspection. A controlled within-subjects user study compares the AR-assisted inspection approach with a traditional inspection method using the SUS score, providing empirical evidence of perceived usability and user acceptance. As described in Section III-B, all participants completed the same railway wheel inspection task using both a traditional inspection approach and the AR-based system, enabling direct within-subject comparison of perceived usability.

Each participant completed the SUS questionnaire immediately after finishing each inspection condition. The same set of SUS items was used for both conditions, with wording adapted only to reference the respective inspection approach. This evaluation approach allowed usability perceptions to be assessed independently for each method while minimizing the influence of individual differences.

The AR-based inspection approach, as an instantiation of the proposed framework, achieved a mean SUS score of 52.24 and a standard deviation (SD) of 17.54, while the traditional inspection approach achieved a mean SUS score of 58.55 and an SD of 18.07. A paired comparison of SUS scores did not reveal a statistically significant difference between the two inspection methods (paired t-test: $t = -1.087$, $p = 0.291$). Median SUS scores were comparable across conditions, with AR = 57.5 and the Traditional approach = 55.0, indicating substantial overlap in perceived usability between the two approaches.

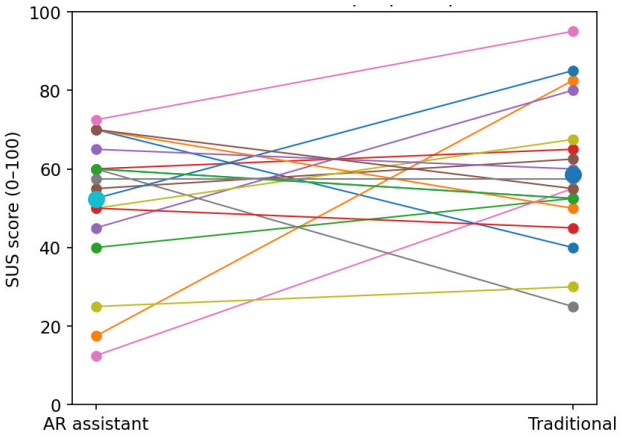


FIGURE 7. Paired SUS scores for individual participants comparing the AR-based and traditional inspection approaches.

Both inspection approaches scored within the marginal usability range (50–68), indicating that neither method was perceived as highly usable under the evaluated conditions. While the traditional approach achieved a slightly higher mean SUS score, comparable median values and overlapping score distributions suggest that participants perceived the usability of the two approaches similarly. Rather than evidence of satisfactory usability, these results represent a baseline for the proposed framework instantiation, indicating

that the AR-based approach did not introduce additional usability barriers despite being novel to most participants. The relatively large standard deviations in both conditions reflect considerable inter-participant variability, likely attributable to differences in prior inspection experience, familiarity with digital tools, comfort with multimodal interaction, and a language mismatch between participants' native language and the English-language interface. Figure 7 shows paired SUS scores for individual participants, highlighting within-subject differences between the two approaches. These findings motivate further refinement of both the system design and evaluation approach in future work.

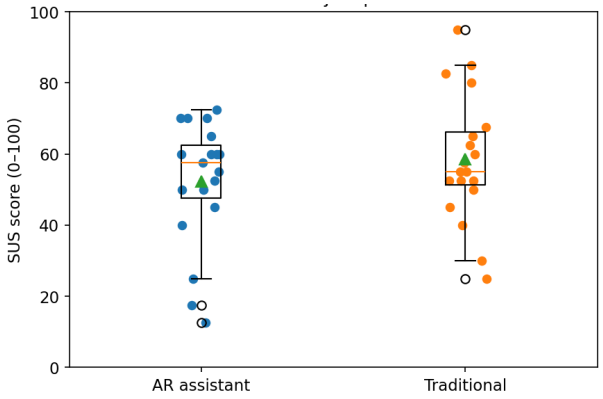


FIGURE 8. Distribution of SUS scores for the AR-based and traditional inspection approaches.

Figure 8 presents the distribution of SUS scores for both inspection methods. Median values and interquartile ranges indicate comparable usability across conditions, with considerable variability observed in both cases.

In addition to the overall usability analysis, an exploratory preference-based examination was conducted to identify candidate usability dimensions for future investigation. Given the limited sample size of this subset, no statistical inference should be drawn from item-level comparisons. This analysis is presented as contextual insight only and focuses on understanding which usability aspects may have contributed to this preference. Figure 9 presents mean item-level SUS scores for this subset of participants.

Participants who preferred the AR-based inspection approach rated the system higher in all aspects. Some aspects exhibited larger differences compared to the traditional method, particularly on SUS items Q1, Q2, Q4, Q5, Q7, Q9, and Q10. While the differences for the remaining items, Q3, Q6, and Q8, are comparatively smaller. Items exhibiting larger differences are primarily associated with adoption intent, confidence, learnability, functionality integration, and perceived system support.

In contrast, items related to general ease of use and consistency show relatively smaller differences between the two approaches. This indicates that baseline usability perceptions for these aspects were similar for both the AR-based and traditional inspection methods within this participant subgroup.

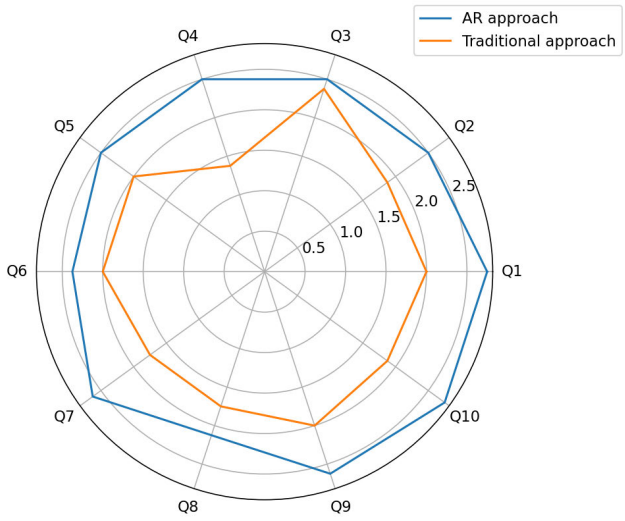


FIGURE 9. Radar plot showing mean SUS item scores for the AR-based and traditional inspection approaches, calculated only for participants expressing a preference for the AR-based system.

To further explore factors influencing user preference, an additional analysis examined the relationship between prior AR familiarity and stated inspection approach preference. Participants were grouped based on whether they reported no prior familiarity with AR technologies or some level of prior familiarity. Figure 10 illustrates the distribution of inspection approach preferences within these two groups. Among participants with no prior AR familiarity, approximately 70% preferred the traditional inspection approach, while around 30% preferred the AR-based approach. In contrast, among participants with some prior AR familiarity, approximately 55% preferred the AR-based inspection approach, around 10% reported no clear preference, and approximately 35% preferred the traditional method.

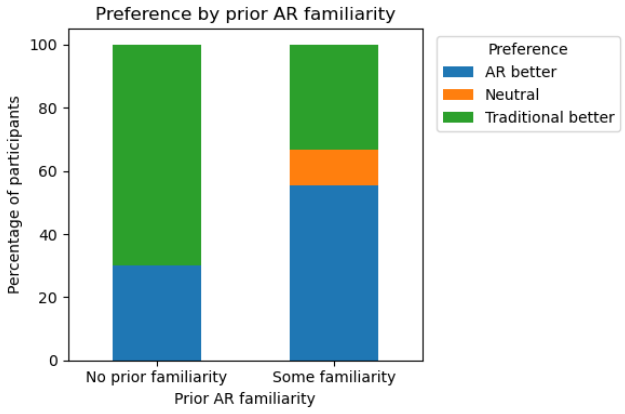


FIGURE 10. Distribution of inspection approach preferences grouped by prior AR familiarity.

Participants with no prior AR familiarity more frequently preferred the traditional inspection approach, whereas participants with some prior AR familiarity showed a higher

tendency to prefer the AR-based inspection approach. This trend suggests that familiarity with AR interaction paradigms may influence user acceptance and perceived suitability of AR-supported inspection workflows. Given the limited sample size, this analysis is exploratory and intended to provide contextual insight rather than statistical generalization.

V. LIMITATIONS AND FUTURE WORK

Some limitations should be acknowledged within this study. The user study involved a relatively limited number of participants and was conducted within a controlled experimental setting. While this approach enabled direct within-subject comparison between inspection methods, it may not fully capture long-term usability perceptions or real-world operational constraints encountered during routine railway maintenance. The within-subjects design, while appropriate for controlling individual differences, provides moderate statistical power for detecting medium-to-large effect sizes with 19 participants. In addition, the participant group consisted primarily of early-career personnel with limited railway maintenance experience. Findings may therefore not directly generalise to more experienced maintenance technicians, who may have different interaction preferences, procedural familiarity, and cognitive task demands.

The evaluation relied solely on the SUS as the primary usability instrument. While SUS is a well-validated measure of perceived usability, it does not capture objective performance indicators such as task completion time, inspection error rate, or the number of interaction events. For a system designed to reduce cognitive burden and support decision-making during safety-critical inspection tasks, these dimensions are particularly relevant. Future evaluations would incorporate complementary tools such as the NASA Task Load Index to assess cognitive load, alongside objective task performance metrics, to provide a more comprehensive assessment of the system's operational benefits.

A further limitation concerns the language of the study materials. While the AR-based maintenance assistant and all study materials were presented in English, the participants were native Swedish speakers. Although participants reported being comfortable using English, some indicated that interaction in their native language would have been preferable. This language mismatch may have influenced usability perceptions, particularly for cognitive and confidence-related aspects of interaction. Future implementations should therefore support multilingual interfaces to better accommodate diverse user populations and reduce potential language-related cognitive load.

The current instantiation of the proposed framework generates inspection summaries locally and does not integrate with broader maintenance information systems. Future work could extend the framework by exploring integration with external maintenance and asset management platforms, for example, through cloud-based data synchronization. Such integration would enable inspection logs to be automatically uploaded for centralized storage, further analysis, and alignment with existing maintenance management workflows, as demonstrated in related work [18]. In addition, future framework extensions may support adaptive assistance strategies, such as adjustable guidance levels, customizable information density, and optional interaction modalities, to better accommodate diverse user needs and improve acceptance in real-world maintenance environments. Future work could also address deployment challenges identified in this study, including the impact of variable lighting conditions and high ambient noise levels typical of operational railway environments, which were not evaluated in the controlled laboratory setting.

Finally, this work focuses primarily on maintenance technicians as the intended users of the proposed system. This scope was deliberate, as technicians are directly responsible for executing inspection tasks and are most exposed to the cognitive, procedural, and situational challenges addressed in this study. Other stakeholders, such as maintenance supervisors, planners, and asset managers, play important roles in the broader maintenance process but typically interact with inspection information before or after task execution rather than during it. The structured inspection data and maintenance logs generated by the proposed system could, however, support these additional stakeholders through complementary interfaces for review, coordination, planning, and decision-making. Extending the system to support multiple stakeholder roles and information needs, therefore, represents an important direction for future work.

VI. DISCUSSION

This study proposes a human-centric AR maintenance assistance framework and demonstrates its instantiation through a railway wheel inspection case. Although the framework was instantiated and evaluated within a specific inspection context, its functional structure is not limited to this asset or domain. The identified inspection task demands, such as procedurally constrained workflows, cognitive load, decision uncertainty, and documentation requirements, are common across safety-critical inspection and maintenance activities. As such, the proposed framework provides transferable design knowledge that may be adapted to other railway assets or inspection tasks in different industrial domains, provided that domain-specific procedures and decision-support knowledge are incorporated. Potential application scenarios may include overhead line equipment inspection, bogie and axle maintenance, track geometry assessment, and safety-critical inspection tasks in related industries such as aviation, mining, and construction.

The AR-based maintenance assistant developed in this study illustrates how the framework can be realized in practice. The instantiation integrates guided task execution, contextual decision support, and structured documentation within a single inspection workflow. Participants who preferred the AR-based inspection approach reported higher ratings related to learnability, confidence, and functional integration, suggesting that the consolidation of multiple support functions can enhance perceived task support. Although decision accuracy was not directly measured, the

system is designed to support technicians in consistently applying inspection standards and interpreting findings within context, which aligns with the framework's emphasis on assistive intelligence that augments rather than replaces human judgment, consistent with Industry 5.0 principles.

From an evaluation perspective, both inspection approaches scored within the marginal usability range, indicating that neither method was perceived as highly usable under the evaluated conditions. The non-significant difference between conditions ($t = -1.087$, $p = 0.291$) should be interpreted cautiously; the observed effect size was small (Cohen's $d = -0.249$), and the study was not sufficiently powered to detect small effects with 19 participants. The absence of a statistically significant difference, therefore, does not constitute evidence of equivalence, but rather reflects the limitations of the study's statistical power. The comparable median scores and overlapping distributions suggest that the AR-based approach did not introduce additional usability barriers relative to the traditional method, despite the novelty of the interaction paradigm. Given the established nature of traditional inspection workflows, this represents a meaningful starting point for iterative development of the AR system rather than confirmation of satisfactory usability for either approach.

The considerable inter-participant variability observed in both conditions, likely reflecting differences in prior inspection experience, AR familiarity, and comfort with multimodal interaction, suggests that a one-size-fits-all AR interface may not be optimal. This points to the need for adaptive assistance strategies that accommodate varying user needs and experience levels.

It should also be noted that the evaluation relied solely on the SUS, which does not capture objective performance indicators or cognitive load. As discussed in the Limitations section, future evaluations should incorporate broader measurement instruments for a more comprehensive assessment.

Overall, the findings suggest that the adoption potential of AR-based maintenance assistance systems depends not only on overall usability but also on alignment with user expectations, experience levels, and task demands. The absence of additional usability barriers relative to traditional methods appears to be a necessary condition for adoption, while perceived benefits in guidance, confidence, and workflow integration play a critical role in shaping user acceptance. This highlights the importance of context-aware and user-adaptive AR system design for successful deployment in industrial maintenance environments.

VII. CONCLUSION

This paper presented a human-centric AR maintenance assistance framework for safety-critical railway inspection tasks, grounded in Industry 5.0 principles and developed following a DSR approach. The framework defines functional elements and information flows that structure inspection support around guided task execution, contextual decision support, and integrated documentation. To illustrate its applicability, the framework was instantiated as an AR-based maintenance assistant for railway wheel inspection.

The usability evaluation showed that both the AR-based and traditional inspection approaches scored within the marginal usability range, with no statistically significant difference between conditions. The results should be understood as a usability baseline for the proposed framework instantiation rather than evidence of satisfactory usability for either approach. Exploratory preference-based analysis indicated that participants favouring the AR approach perceived benefits in functional integration, learnability, confidence, and intention to use, suggesting directions for iterative system improvement. These findings indicate that the proposed framework can be instantiated without introducing additional usability barriers relative to established inspection workflows, while motivating further refinement and evaluation in operational settings.

Beyond usability, the results suggest that AR-based maintenance assistance systems grounded in the proposed framework can support inspection tasks by promoting structured execution, reducing reliance on memory, and facilitating access to relevant information during task execution, while preserving the central role of human judgment. Future work should address the identified deployment considerations, including ergonomic comfort, environmental robustness, and multilingual support, and evaluate the framework across broader operational contexts and user populations to establish its generalisability and long-term adoption potential.

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